CHAPTER C-05

UNDERWRITING AND RATING

Chapter Introduction

In this chapter you will learn the basics of underwriting and rating. You will learn about the different methods of dealing with hazards in the process of rating of risks. You will be able to appreciate the common aspects of underwriting, product approval and rating.

Learning Outcomes

A. Basics of Underwriting

B. Product Filing with IRDAI

C. Basics of Ratemaking

D. Rating factors

After studying this chapter, you should be able to:

1. Define the basics of underwriting
2. Understand the basics of product approvals in India
3. Appreciate rating factors and the importance of ratemaking

A. Basics of Underwriting

In the previous chapters, we have seen that the concept of insurance involves managing risk through pooling. Insurers create a pool consisting of premiums that are made by several individuals/ commercial/ industrial firms/ organizations.

This process of understanding risks, classifying risks, identifying which category they fall into, **deciding whether to accept the risk or not** and if so, how much premium the insurer would require to accept the risk and whether any extra conditions are to be imposed on the risk - all these are part of **underwriting**.

It is also important to know what rate is to be charged and how the rates are made.

Definition

Underwriting is the process of determining whether a risk offered for insurance is acceptable, and if so, at what rates, terms and conditions.

Underwriting comprises the following steps:

i. Assessment and evaluation of hazard and risk in terms of frequency and severity of loss

ii. Formulation of policy coverage and terms and conditions

iii. Fixing of rates of premium

The underwriter decides on whether or not to accept the risk

The next step would be to decide the **rates, terms and conditions** under which the risk is to be accepted.

Underwriting skills are acquired through a continuous learning process involving adequate training, field exposure and deep insights. To be a fire insurance underwriter one needs to have a good knowledge of the likely causes of fire, impact of fire on various physical goods and property, the process involved in an industry, geography, climatic conditions etc.

Similarly a marine insurance underwriter must be aware about port/ road conditions, problems encountered by cargo/ goods in transit or storage, ships and their seaworthiness and so on.

A health underwriter needs to understand the risk profile of the insured, age, medical aspects, fitness levels and family history and measure the effect of each factor affecting the risk.

Sources of information for underwriting

The first stage in any numerical (or statistical) analysis is the collection of data. When pricing a risk, an underwriter should gather as much information as possible to aid accurate assessment.

Sources of information are:

i. **Proposal form or underwriting presentation**

ii. **Risk surveys**

iii. **Historic claims experience data:** For some classes of business, such as personal and motor lines, underwriters often utilise historic claims experience data to provide an indication of the likely future claims experience, and to arrive at a suitable premium.

Underwriting, equity and business sustainability

The need for careful underwriting and risk classification in insurance arises from the simple fact that **all risks are not equal**. Each risk thus needs to be appropriately assessed and priced in accordance with the likelihood of loss occurrence and severity.

Since all risks are not equal, it would not be proper to ask all those who are to be insured, to pay equal premium. **The purpose of underwriting is to classify risks so that, depending on their characteristics and degree of risk posed, an appropriate rate of premium may be charged.** It is important for the underwriter to ensure that the risk evaluation is done properly and the premium charged is neither too low to cover the risk nor too high to make it non-competitive.

The main features of underwriting are as follows

i. To **identify risk** based upon the characteristics

ii. To **determine the level** of risk presented by the proposer

The objectives of underwriting are achieved, in short, by deciding the level of acceptability, adequacy of premium and other terms.

B. Product Filing with IRDAI

Every Insurance product needs to be filed with IRDAI for approval before it is offered for sale. IRDAI allots a Unique Identification number (UIN) for every insurance product. Once products are introduced in the market, there are guidelines to be followed for withdrawing the product as well.

1. The Regulator asks for a clear commitment by the Board of the insurer that it is willing to accept the risks in the policy and agrees to pay the claims. It also asks the insurer to commit that the policy wordings are fair to the customer and that the prices are decided on a scientific basis.

2. The insurer should plan for the possibility of withdrawal of the products in the future and the options that would be available to the policyholder on such withdrawal of the product.

3. The withdrawn product shall not be offered to the prospective customers.

C. Basics of Ratemaking

Insurance is based on transfer of risk to the insurer. By purchasing an insurance policy, the insured is able to reduce the impact of financial losses arising from the peril against which the property is insured. The Insurer needs to adopt a process of calculating a price to cover the future cost of insurance claims and expenses, including a margin for profit. This is known as **ratemaking.**

A rate is the price of a given unit of insurance. For example, a rate may be expressed as Rs.1.00 per mile (per thousand) sum assured for earthquake coverage. Each rate is established after looking at past trends and changes in the current environment that may affect potential losses in the future.

Note that rates are not the same as premiums.

Premium = (Sum Insured) x (rate)

Example

Taking an example of health insurance, numerical or percentage assessments are made on each component of the risk. Factors like age, race, occupation, habits etc. are examined and scored numerically based on predetermined criteria.

The amount of premium to be paid by each depends on a rate, which is determined by two factors;

- ✓ The probability of loss due to a loss event (caused by an insured peril) and
- ✓ The estimated amount of loss that may arise due to the loss event

Example

Assume the average amount of a house being destroyed by fire is Rs 1,00,000.

The probability of the loss of a house being destroyed by fire 1 out of 100 [or 0.01]. That is, the experience is that out of a 100 insured houses, one house gets destroyed by fire.

The expected average loss would be Rs.1,00,000 x 0.01 = Rs. 1000.

So, Insurers would need to charge a minimum of Rs.1000 to insure a house of Rs.1,00,000 value.

How can the insurer ensure that the pool is sufficient to compensate for the losses that are actually incurred?

As seen earlier, the whole mechanism of insurance involves pooling of many similar risks so that the probability of the number of losses (frequency) as well as the extent of loss (severity) becomes predictable. This principle, referred to as 'the law of large numbers' states that as the sample size grows, the results come closer to the expected value. Insurance companies need to sell more policies to more and more people to make their expectations/ predictions work.

An example is that if a coin is tossed, the chances of getting 'heads' or 'tails' is 50:50. However, if the coin is tossed only once, the result can be 100% heads and 0% 'tails' or 0% 'heads' and or 100% tails. However, if one tosses a coin many times, the chance of the average count of 'heads' and 'tails' being 100% and 0% reduces and will get closer to 50:50.

Example

In the field of property insurance, the chances of a wooden structure catching fire are more than stone structures; hence, a higher premium is required to insure the wooden structure.

The same concept applies to Life and Health Insurance also. An individual suffering from high blood pressure or diabetes has higher chances of suffering a heart attack.

Test Yourself 1

Identify the two factors that affect insurance ratemaking.

I. Probability and severity of risk
II. Source and nature of risk
III. Source and timing of risk
IV. Nature and impact of risk

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1. Determining the rate of premium

The pure rate of premium is arrived at on the basis of past loss experience. Therefore, statistical data regarding past losses is most essential for purposes of calculating rates. To fix the rates, it is necessary to give a 'mathematical value' to the risks.

Example

If loss experience of a large number of motor cycles is collected for a period of say 10 years, we will get the sum total of the losses resulting from damage to the vehicles. By expressing this amount of loss as percentage of the total value of motor cycles we can fix the 'mathematical value' of the risk. This may be expressed in the formula given below:

$M = \frac{L \times 100}{V}$	L refers to the sum total of the losses and V to the total values of all the motor cycles and M to average loss percentage.

Let us suppose that:

- ✓ The Value of a motor cycle: Rs. 50,000/ -
- ✓ Loss experience: Out of 1000 motor cycles, 50 motor cycles get stolen over 10 years
- ✓ On an average, 5 motor cycles become total losses due to theft every year

Applying the formula, the result will be:

Losses per year (Rs. 50,000 X 5) = Rs. 2,50,000

Total Values of 1000 motor vehicles (Rs. 50,000 X 1000) **= Rs. 5,00,00,000**

This means that average loss percentage per vehicle (L/ V) x 100= [2,50,000/ 5,00,00,000] x 100 = 0.5%

Therefore the rate of premium that a motor cycle owner pays is half a percent of Rs. 50,000/ - i.e. Rs. 250/ - per year. This is called the **'Pure' premium,** also known as 'Burning Cost'.

At the rate of Rs. 250 per motor cycle, Rs. 2.5 lakhs is collected which is paid out in claims on total losses of 5 vehicles.

If the pure premium, which is arrived above, is collected it would constitute a fund which will be sufficient only to pay for losses.

In the example above we can see that there is no surplus. But insurance operations also involve costs of administration (expenses of management) and costs of procurement of business (agency commission). It is also necessary to provide a margin for unexpected heavy losses.

Finally, since insurance is transacted on a commercial basis, like any other business, it is necessary to provide for a margin of profit which is a return on the capital invested in the business.

Therefore, the 'pure premium' is suitably loaded or increased by adding percentages to provide for expenses, reserves and profits.

The final rate of premium will consist of the following components:

- ✓ Loss payments
- ✓ Loss expenses (e.g. survey fees)
- ✓ Agency commission
- ✓ Expenses of management
- ✓ Margin for reserves for unexpected heavy losses e.g. 7 total losses against 5 expected
- ✓ Margin for profits

By taking all the relevant rating factors into consideration, one can ensure the rates are adequate, excessive or unfairly discriminatory as between risks of similar type and quality.

Test Yourself 2

What is pure premium?

I. Premium sufficiently big enough to pay for losses only
II. Premium applicable to marginal members of the society
III. Premium after loading for administrative costs
IV. Premium derived from the most recent loss experience period

2. Deductible

'Deductible' or 'excess' is a cost-sharing provision between an insurer and insured. Deductibles provide that only the claims in excess of a particular threshold are payable by the insurer. In other words, the insurer will not be liable for claims below a specified level. The level or the threshold would be set as a fixed amount, or a percentage or even as a specified period of time (when it is called time-excess.) In case of health policies, there could be a condition that claims would be payable only if the hospitalization is beyond a specified number of days/ hours. Deductibles are not used in life policies.

In products such as property, motor and home insurances, deductibles are predetermined amounts that the insured must bear towards an indemnity claim. Deductibles can be compulsory for some policies or voluntary. Insurers generally charge lower premiums when the insured voluntarily opt for higher deductibles. An agent must examine how specific deductibles work and inform the insured whether the deductible is applicable on a 'per year' or 'per event' basis.

There are various reasons for having deductibles. Corporate customers covering factories, multiple cargo consignments, large groups of employee, public liability exposures etc. and having huge amounts of Sum Insured, may prefer to bear small claims themselves and avoid the documentation to prove claims. For example, a large factory owner paying lakhs or rupees as premium may not be bothered about a minor repair cost of a machine amounting to around Rs.2,000.

Some type of policies may need the insured also to bear some part of the loss to ensure that he/ she takes due care. For instance, health insurers may insist on a deductible so that insured would not overspend on costly hospital rooms just because insurance is there. Some Insurers also may not prefer spending time on processing small claims. Also, in certain situations, insurers may not want to get exposed to the financial stress caused by accumulation of a large number of small losses at one location. For example, a small flood in an industrial estate area can cause many low value claims from all the warehouses in the area.

Franchise: Franchise refers to a threshold set, usually as a percentage of the sum insured, below which no claim is admissible, as in the case of deductibles. However, when the claim amount is beyond the franchise limit, the entire claim is admissible by the insurer. In other words, franchise determines the minimum threshold of the insurance companies' financial responsibility. Franchise will apply to the policy in the same way and for the same reasons as a deductible in case of claims below the threshold, but in the event of a claim exceeding the franchise, the full amount of the loss will be paid.

D. Rating factors

The relevant elements that are used to add up the rates and make the rating plan are referred to as **rating factors**. Insurers use 'rating factors' to determine the risk and to decide the price they will charge.

- ✓ The Insurer uses his assessments to establish a base rate.
- ✓ The Insurer then adjusts this rate with discounts applied for positive features such as superior fire protection on property risk and loadings applied for adverse features such as presence of inflammable materials in the premises.
- ✓ In Life Insurance the usual practice is to apply loading for adverse health, habits, heredity or occupational factors.

Key Terms

- Deductibles
- Franchise

Answers to Test Yourself

Answer 1 - The correct option is I.

Answer 2 - The correct option is I.